Impact of Quick Commerce on Retail

Team Members -

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Plan of Action

1. Introduction
2. Objective
3. Data Collection
4. EDA
5. Hypothesis Testing
6. Model Training
7. Analysis
8. Conclusion

Introduction

- The retail sector is one of the most important parts of the Indian economy, providing goods and services to consumers through both organized and unorganized formats.
- In recent years, Quick Commerce (Q-commerce) has emerged as a new retail model that focuses on delivering daily-need products such as groceries, medicines, and household items within 10–30 minutes using technology, hyperlocal fulfillment centers, and fast delivery networks.
- It has changed consumer buying behavior, increased expectations for speed and convenience, and created strong competition for local and organized retail stores.
- This study examines the impact of quick commerce on the retail sector, focusing on its effects on consumer behavior, competitive dynamics, and the future of traditional retail businesses.

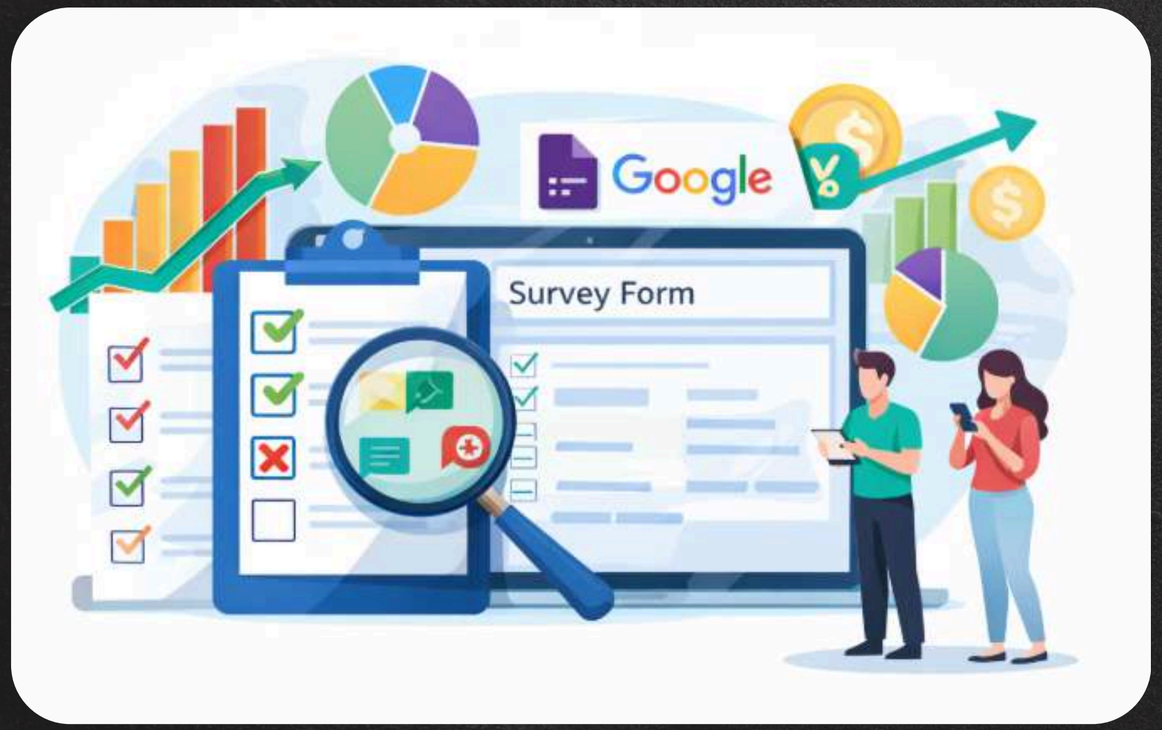
Objective

To analyze the impact of Quick Commerce on Retail stores.

To identify key factors driving customers toward Quick Commerce over traditional retail shopping.

To examine changes in consumer behavior, shopping frequency, and spending patterns due to e-commerce adoption.

Data Collection



Methodology

Google form : Primary Data

Questionnaire

17 questions - Shopping Preference as target column

Tests

One way Anova
Chi Square
Logistic Regression

Exploratory Data Analysis (EDA)

- **The Dataset Contains 19 Columns and 147 Rows.**
- **Out of 19 columns 15 are categorical columns and 4 Numerical columns.**
- **There are no duplicate entries in this dataset.**
- **There are no missing / null values in the dataset.**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 147 entries, 0 to 146
Data columns (total 19 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Age Group                   147 non-null    object
 1   Gender                      147 non-null    object
 2   Shopping_Preference         147 non-null    object
 3   Platform                    147 non-null    object
 4   QC_Frequency                147 non-null    object
 5   Products_Bought             147 non-null    object
 6   Cost_Effective              147 non-null    object
 7   Retail_Frequency            147 non-null    object
 8   Retail_Visit_Reduced        147 non-null    object
 9   Offers_Influence            147 non-null    object
 10  Trust_Quality               147 non-null    object
 11  Retail_Losing_Customers     147 non-null    object
 12  Prefer_Local_Fast_Delivery  147 non-null    object
 13  Replacement_Score           147 non-null    int64
 14  Category_Shift              147 non-null    object
 15  Local_Shops_Closed          147 non-null    object
 16  Shopping_habits             147 non-null    int64
 17  Monthly_Spend               147 non-null    int64
 18  Overall_Satisfaction        147 non-null    int64
dtypes: int64(4), object(15)
memory usage: 21.9+ KB
```

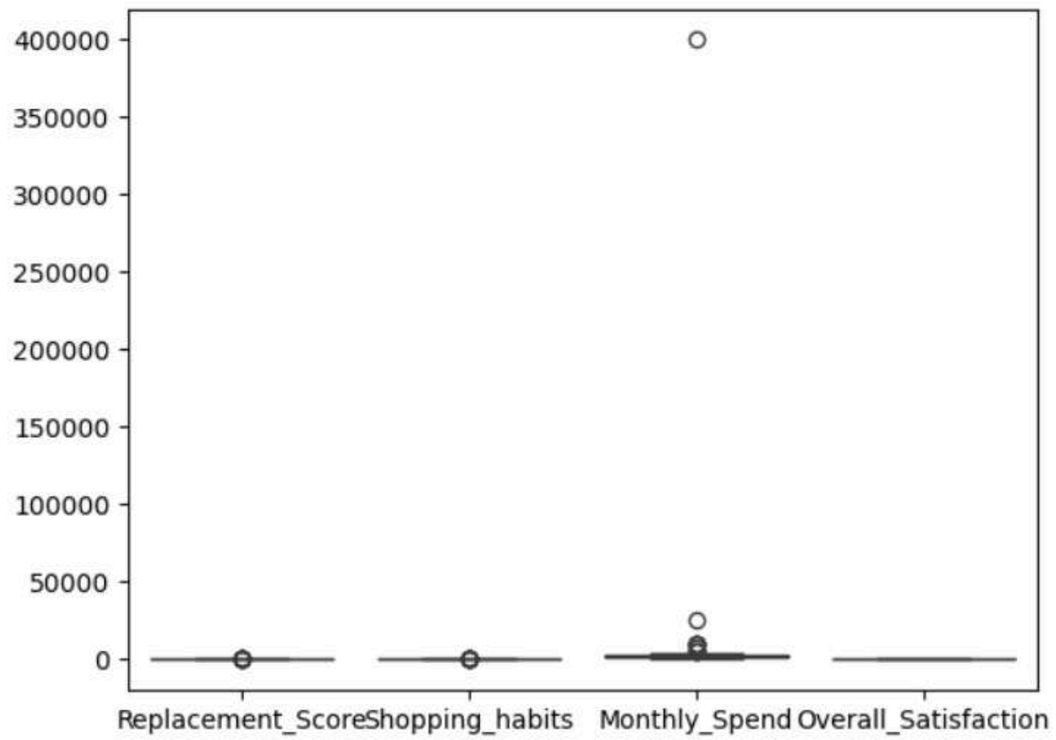
```
Age Group                     0
Gender                        0
Shopping_Preference           0
Platform                      0
QC_Frequency                  0
Products_Bought               0
Cost_Effective                0
Retail_Frequency              0
Retail_Visit_Reduced          0
Offers_Influence              0
Trust_Quality                 0
Retail_Losing_Customers       0
Prefer_Local_Fast_Delivery    0
Replacement_Score             0
Category_Shift                0
Local_Shops_Closed            0
Shopping_habits               0
Monthly_Spend                 0
Overall_Satisfaction          0
dtype: int64
```

Outlier detection & Treatment

Outlier Detection

```
sns.boxplot(data=df)
```

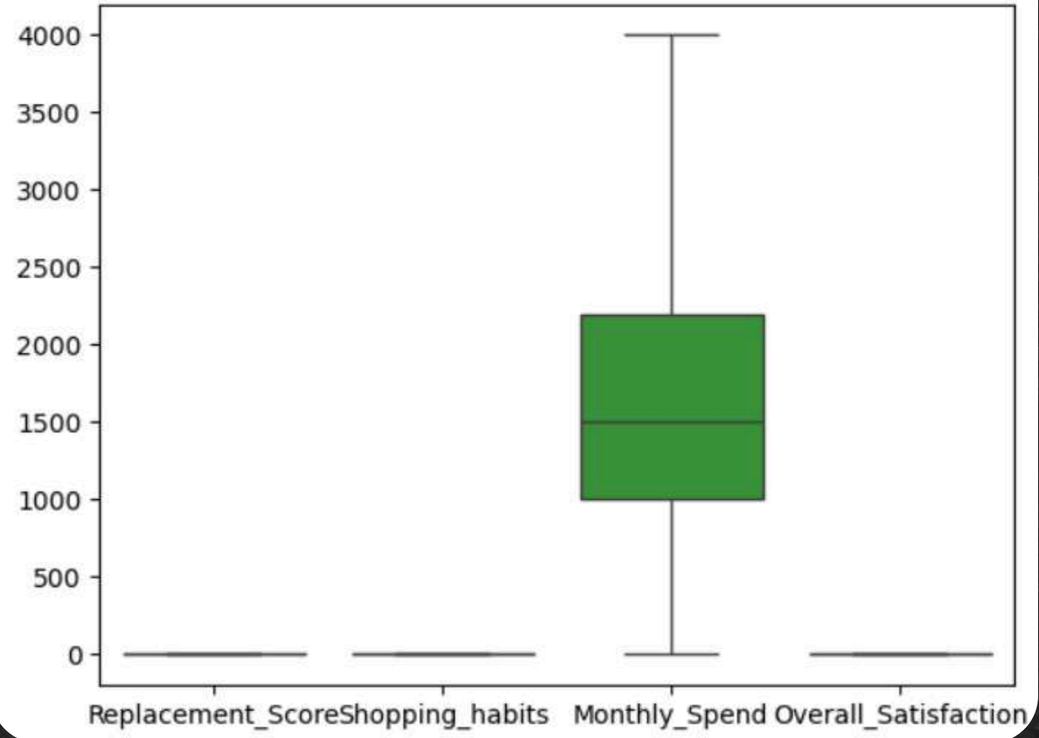
```
<Axes: >
```



Outlier Treatment Using Winsorizing

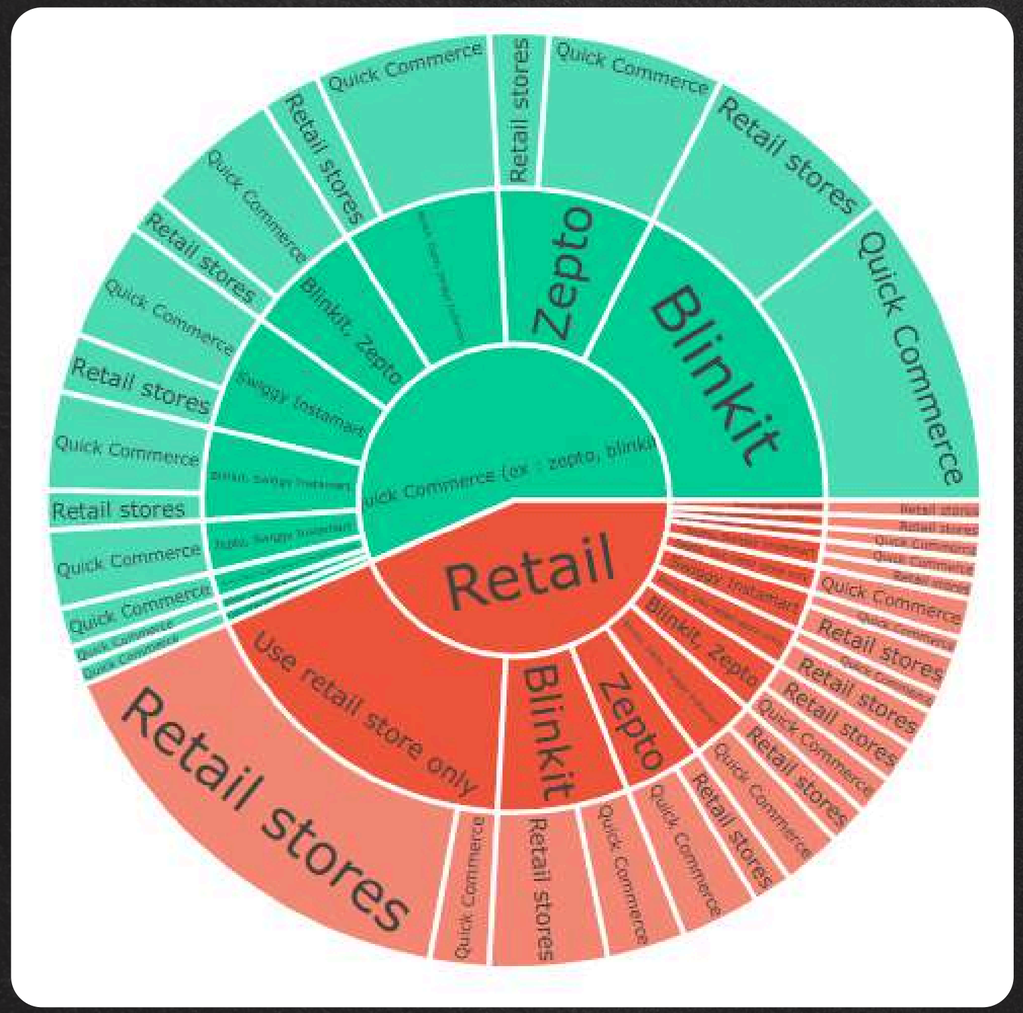
```
sns.boxplot(data=df)
```

```
<Axes: >
```



Hypothesis Testing

Shopping Preference VS Platforms Preferred



The Chi-square test indicates a statistically significant association between shopping preference and platform usage ($p < 0.05$).

We are testing whether a consumer's shopping preference has any association with the platform they prefer to use.

```
Chi-square Statistic: 47.909279333112984
Degrees of Freedom: 11
P-value: 1.4818384979534327e-06
```

Hypothesis Testing

Shopping Preference VS Cost Effectiveness

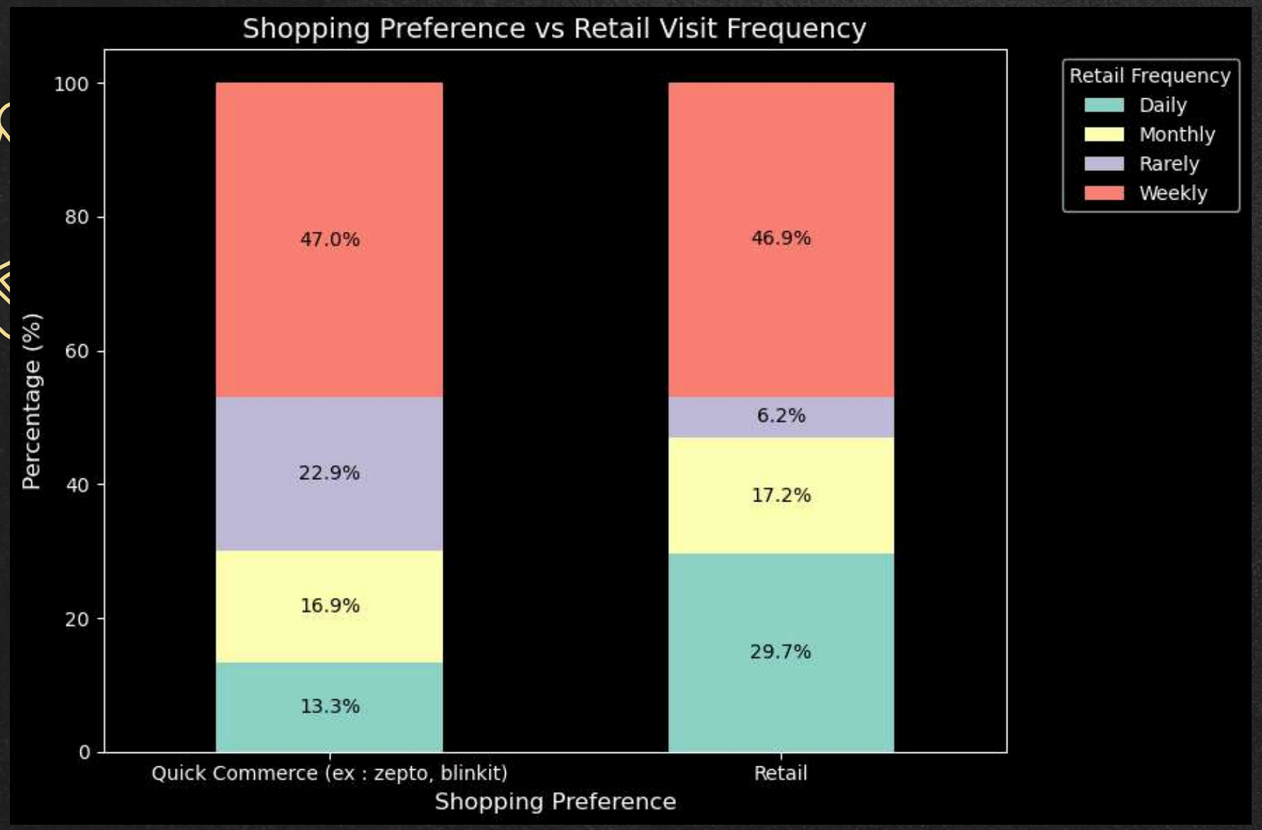
The graph indicates that consumers perceive the shopping channel they prefer as more cost effective.

Highlighting that cost perception plays a decisive role in reinforcing shopping behavior between quick commerce and traditional retail.

```
Chi-square Statistic: 19.550566834057616
Degrees of Freedom: 1
P-value: 9.797166308367832e-06
```

Hypothesis Testing

Shopping Preference VS Retail_Frequency



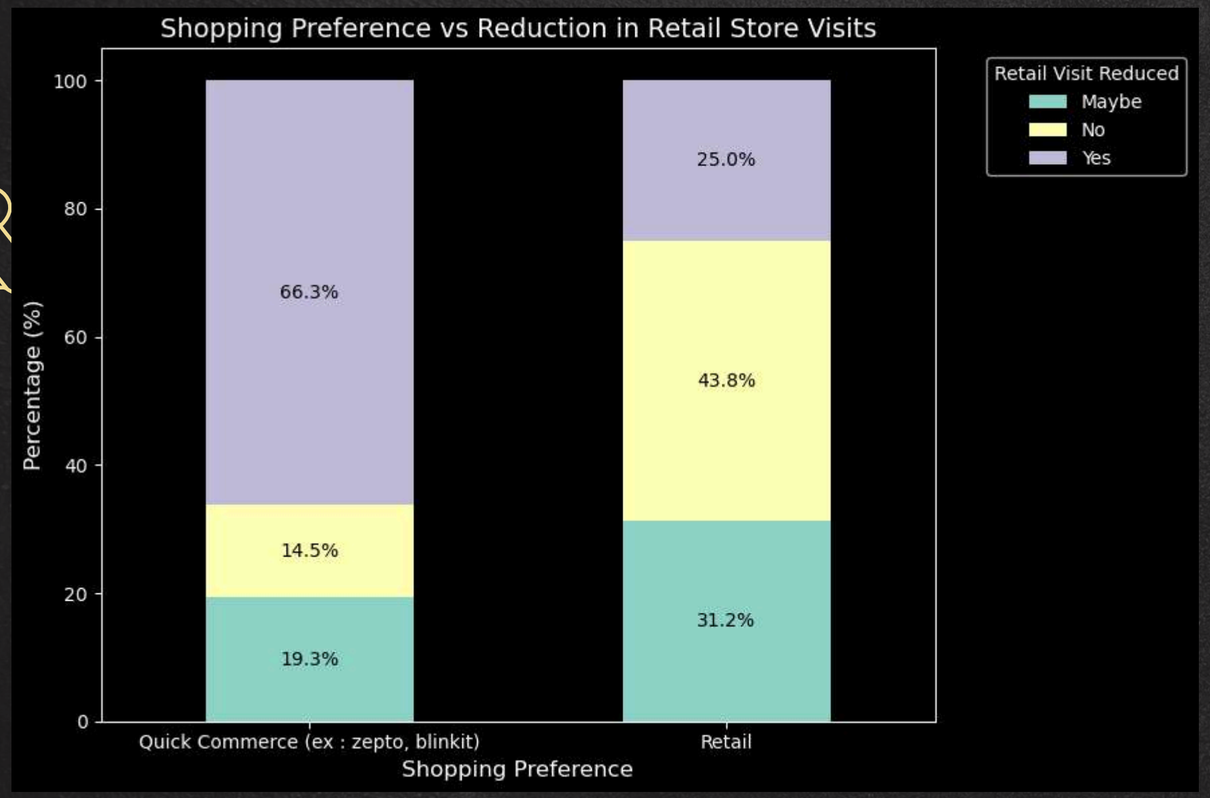
The stacked bar chart shows that consumers who prefer quick commerce tend to visit retail stores less frequently,
while those who prefer retail shopping continue to visit stores more regularly.

This visual evidence supports the Chi-square result, indicating a significant relationship between shopping preference and retail visit frequency

```
Chi-square Statistic: 11.180860234415924
Degrees of Freedom: 3
P-value: 0.010787036559226776
```

Hypothesis Testing

Shopping Preference VS Retail Store Visit Reduced



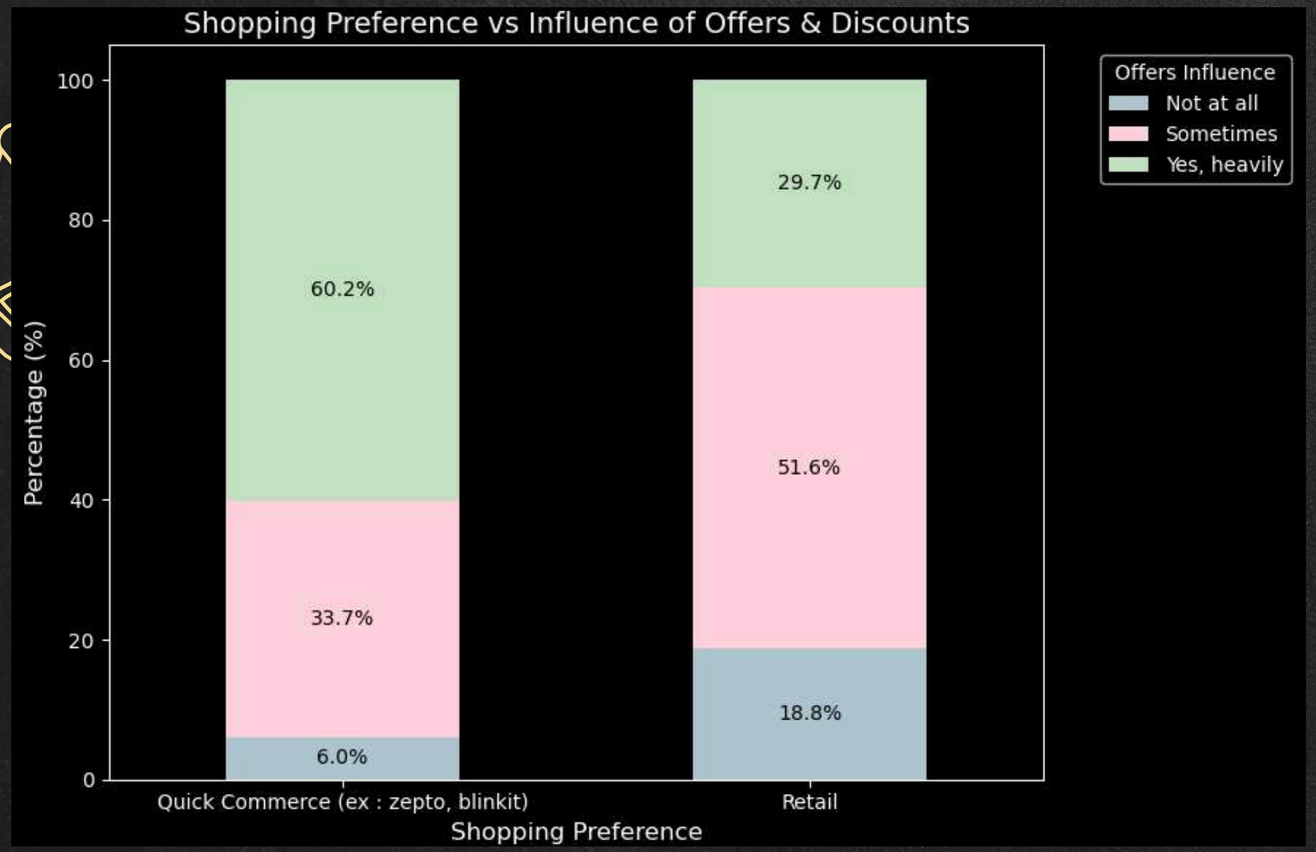
The visualization shows that consumers who prefer quick commerce report a significantly higher reduction in retail store visits compared to retail oriented shoppers.

This supports the Chi-square result and indicates that quick commerce platforms are directly contributing to declining footfall in traditional retail stores.

```
Chi-square Statistic: 11.180860234415924
Degrees of Freedom: 3
P-value: 0.010787036559226776
```

Hypothesis Testing

Shopping Preference VS Influence of Offers & Discount



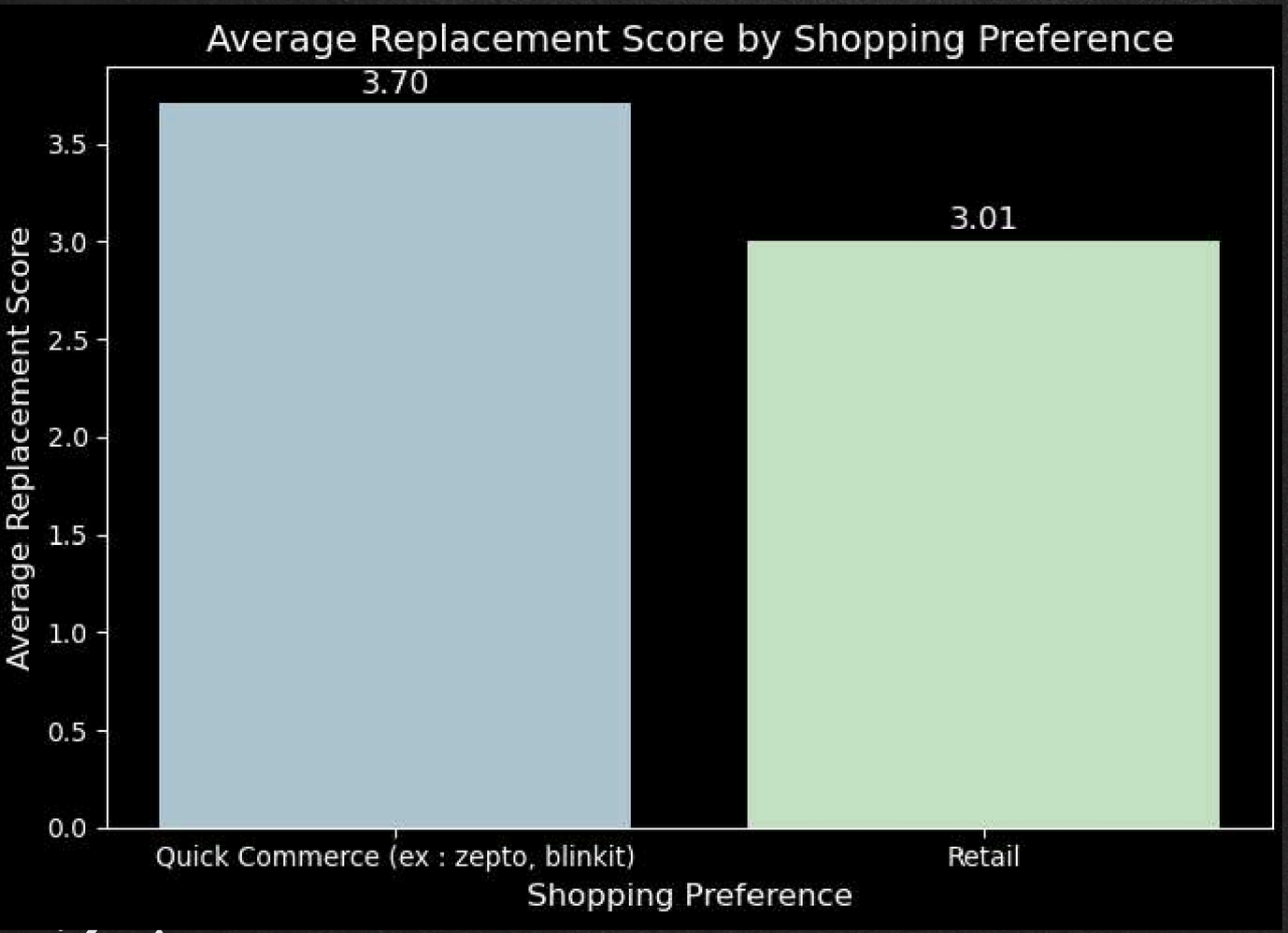
The analysis indicates a significant association between shopping preference and the influence of offers and discounts.

Consumers preferring quick commerce are more strongly driven by promotional offers, highlighting the importance of discount-based strategies in shaping customer behavior.

```
Chi-square Statistic: 15.014779870183006
Degrees of Freedom: 2
P-value: 0.0005490121777034775
```

Hypothesis Testing

Shopping Preference Vs Store Visit Replaced

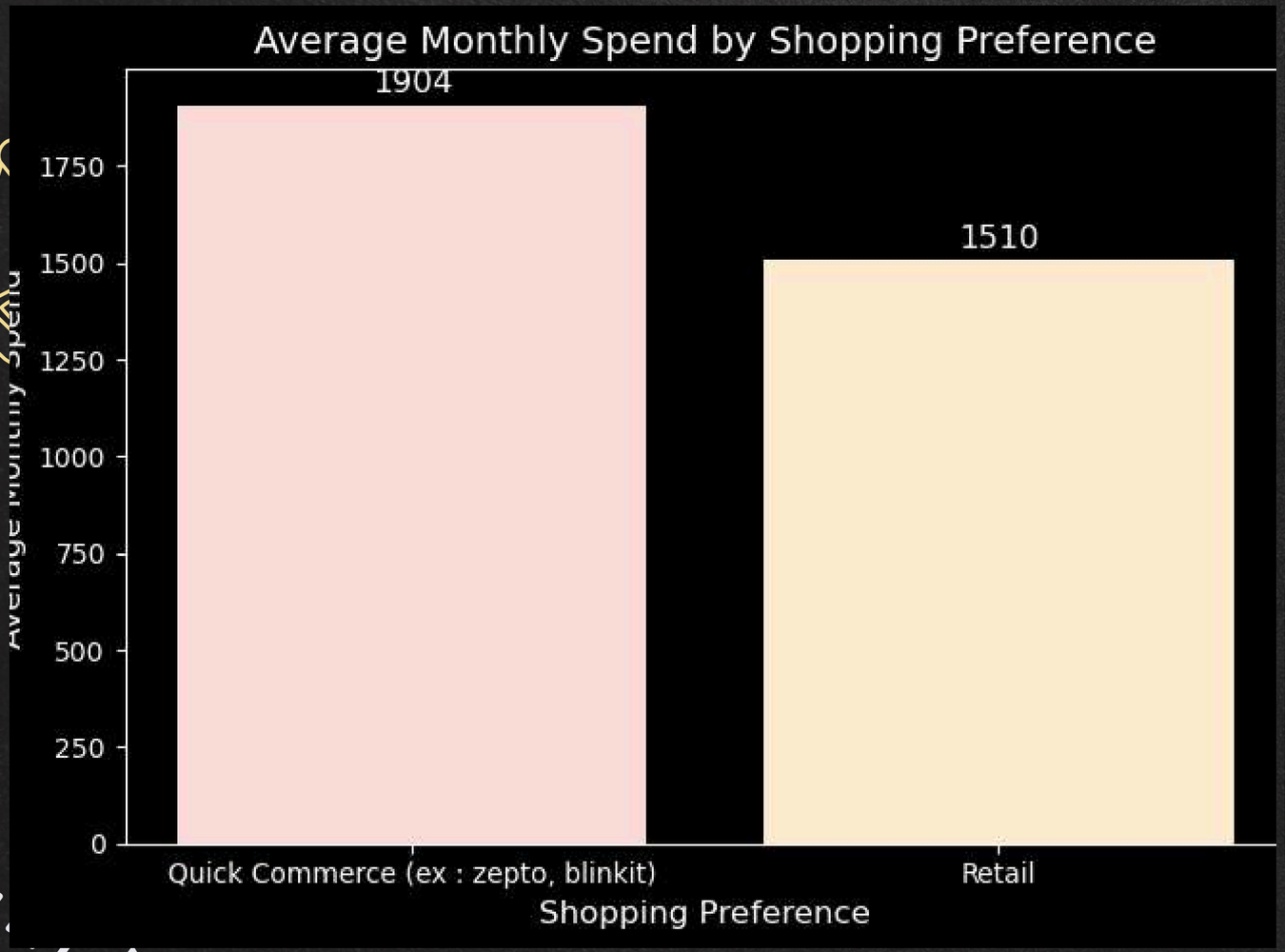


A One - Way Anova Test indicates customers who prefer quick commerce platforms report a higher average replacement score (3.70) compared to retail shoppers (3.01).

```
F-statistic: 19.76913723
P-value: 0.00001726
```

Hypothesis Testing

Shopping Preference VS Monthly Spend



The One way Anova Test analysis indicates a significant association between shopping preference and Monthly spend.

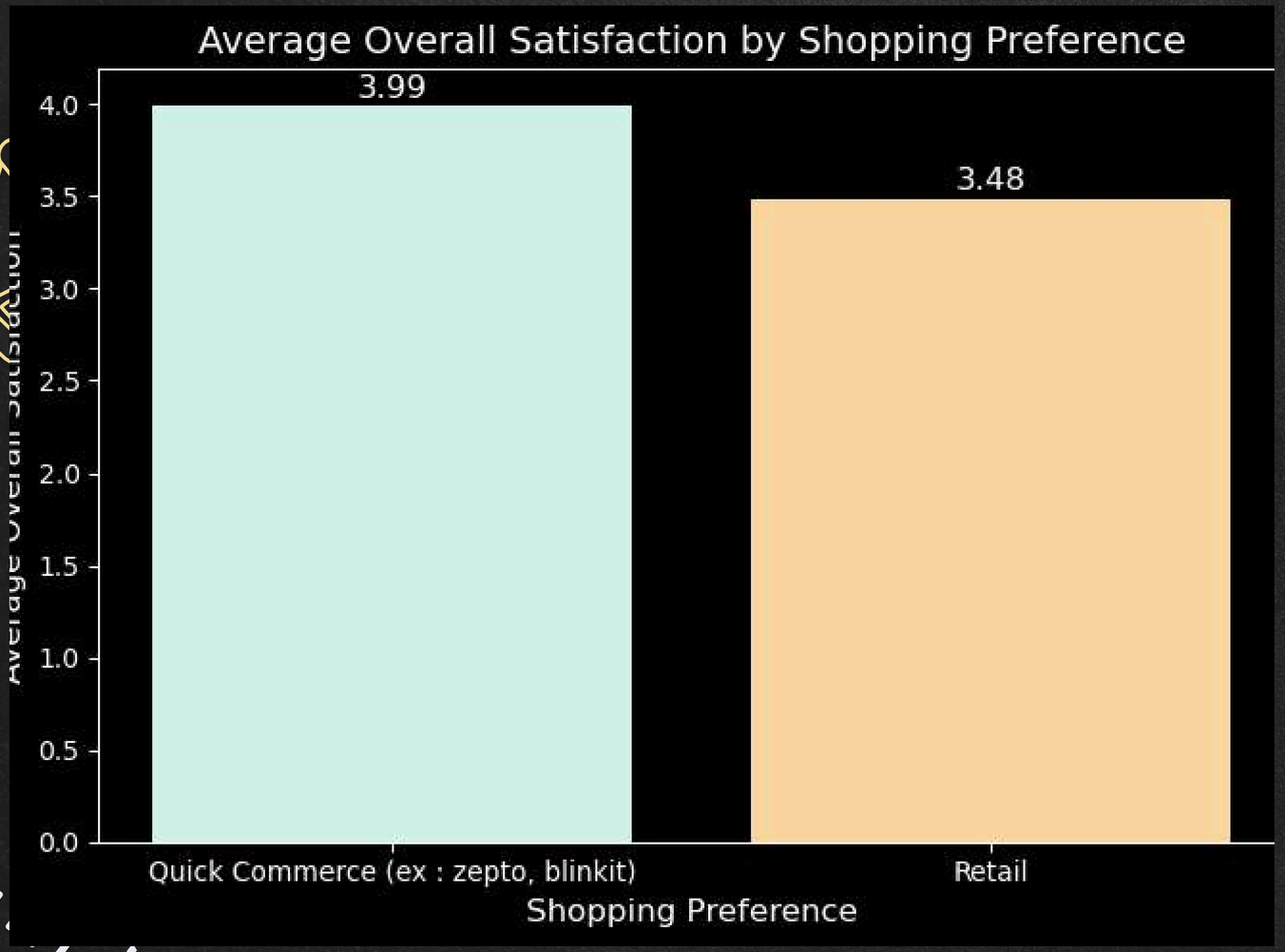
Shopping preference significantly influences monthly spending behavior, with quick commerce emerging as a stronger revenue-generating channel.

F-statistic: 5.5649

P-value: 0.0197

Hypothesis Testing

Shopping Preference VS Overall Satisfaction for Quick commerce



The ANOVA results show a statistically significant difference in overall satisfaction between quick commerce and retail shoppers.

Customers using quick commerce platforms report higher satisfaction levels, indicating superior performance in convenience, delivery speed, and service experience

F-statistic: 8.80121895
P-value: 0.00352196

Model Training Using Logistic Regression

Step 1 : Parting Qualitative and Quantitative Data

Parting qualitative and quantitative data

```
Quantitative_Variable = df.select_dtypes(include=[np.number])
Quantitative_Variable.head(3)
```

	Replacement_Score	Shopping_habits	Monthly_Spend	Overall_Satisfaction
0	4.0	4.0	1200	4
1	3.0	3.0	2500	5
2	4.0	5.0	1800	4

```
Qualitative_Variable = df.select_dtypes(include=['object'])
Qualitative_Variable.head(3)
```


```
Qualitative_Variable = df.select_dtypes(include=['object'])
Qualitative_Variable.head(3)
```


	Age Group	Gender	Shopping_Preference	Platform	QC_Frequency	Products_Bought	Cost_Effective	Retail_Frequency	Retail_Visit_Reduced	Offers_Influ
0	15–25	Male	Quick Commerce (ex : zepto, blinkit)	Blinkit	2–3 times a week	Snacks & beverages	Quick Commerce	Weekly	Yes	Yes, h
1	26-35	Male	Retail	Blinkit, Swiggy Instamart	Once a week	Groceries	Retail stores	Weekly	Yes	Some
2	15–25	Male	Quick Commerce (ex : zepto, blinkit)	Blinkit, Zepto, Swiggy Instamart	2–3 times a week	Groceries, Snacks & beverages, Fruits & vegeta...	Quick Commerce	Monthly	Maybe	Some

Step 2: Variable Encoding for Qualitative columns

Variable encoding for qualitative cols

[44]:
```
from sklearn.preprocessing import LabelEncoder

Qualitative_Variable = Qualitative_Variable.apply(LabelEncoder().fit_transform) # label in ascending order
Qualitative_Variable.head(3)
```

[44]:

	Age Group	Gender	Shopping_Preference	Platform	QC_Frequency	Products_Bought	Cost_Effective	Retail_Frequency	Retail_Visit_Reduced	Offers_Influence	Trust_Qualit
0	0	1	0	0	0	6	0	3	2	2	
1	1	1	1	1	3	2	1	3	2	1	
2	0	1	0	4	0	5	0	1	0	1	

Model Training Using Logistic Regression

Step 3 : Merging Qualitative and Quantitative Columns

Concating number and encoded

```
[45]: Final_combined = pd.concat([Quantitative_Variable,Qualitative_Variable],axis=1)
      Final_combined.head()
```

[45]:

	Replacement_Score	Shopping_habits	Monthly_Spend	Overall_Satisfaction	Age Group	Gender	Shopping_Preference	Platform	QC_Frequency	Products_Bought	Cost_Effe
0	4.0	4.0	1200	4	0	1	0	0	0	6	
1	3.0	3.0	2500	5	1	1	1	1	3	2	
2	4.0	5.0	1800	4	0	1	0	4	0	5	
3	4.0	4.0	900	3	0	1	0	0	3	4	
4	4.0	3.0	3200	5	0	1	0	1	3	4	

Step 4: Data Partition for Train - Test Split

Data Partition ¶

```
[46]: from sklearn.model_selection import train_test_split

      X = Final_combined.drop('Shopping_Preference', axis = 1)
      Y = Final_combined[['Shopping_Preference']]

      X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.30, random_state=120)
```

```
[47]: train = pd.concat([X_train,y_train],axis=1)

      print("No.of Columns",train.shape[1])
      train.head()
```

No.of Columns 19

[47]:

	Replacement_Score	Shopping_habits	Monthly_Spend	Overall_Satisfaction	Age Group	Gender	Platform	QC_Frequency	Products_Bought	Cost_Effective	Retail_Frequen
12	3.0	3.0	2000	4	0	1	9	0	6	0	
36	3.0	3.0	3600	5	0	1	8	0	6	1	
86	4.0	4.0	1600	4	0	1	3	0	4	0	
51	4.0	4.0	1800	4	0	1	10	3	5	0	
94	3.0	2.0	450	3	3	0	2	2	6	1	

Model Training Using Logistic Regression

Step 5 : Feature selection for best columns

```
Model3.feature_names_in_
```

```
array(['Replacement_Score', 'Shopping_habits', 'Monthly_Spend',
       'Overall_Satisfaction', 'Age Group', 'Gender', 'Platform',
       'QC_Frequency', 'Products_Bought', 'Cost_Effective',
       'Retail_Frequency', 'Retail_Visit_Reduced', 'Offers_Influence',
       'Trust_Quality', 'Retail_Losing_Customers',
       'Prefer_Local_Fast_Delivery', 'Category_Shift',
       'Local_Shops_Closed'], dtype=object)
```

```
Model3.get_feature_names_out()
```

```
array(['Replacement_Score', 'Monthly_Spend', 'Platform', 'Cost_Effective',
       'Category_Shift'], dtype=object)
```

Step 6: Calculating Beta 0 & Beta 1 values

```
Model2.intercept_  # beto
```

```
array([0.75946365])
```

```
Model2.coef_  # Beta 1 --
```

```
array([[-3.36673347e-01, -2.87594422e-04,  1.92258935e-01,
         1.38462431e+00, -1.01498530e+00]])
```

```
Model2.feature_names_in_
```

```
array(['Replacement_Score', 'Monthly_Spend', 'Platform', 'Cost_Effective',
       'Category_Shift'], dtype=object)
```

Model Training Using Logistic Regression

Step 7 : Model Accuracy

Model Performance Metrics

```
from sklearn.metrics import confusion_matrix
matrix = confusion_matrix(train['Predicted'], train['Shopping_Preference'])
matrix
```

```
array([[49, 23],
       [ 3, 27]], dtype=int64)
```

```
Accuracy_Train=((49+27)/(102)*100)
print(Accuracy_Train)
print('Accuracy of Model is', round(Accuracy_Train,2), 'So, we say Model good fit data')
```

```
74.50980392156863
Accuracy of Model is 74.51 So, we say Model good fit data
```

Step 8: Test Accuracy

Model Performance Metrics on Test data

```
from sklearn.metrics import confusion_matrix
matrix = confusion_matrix(test['Predicted'],test['Shopping_Preference'])
print(matrix)
```

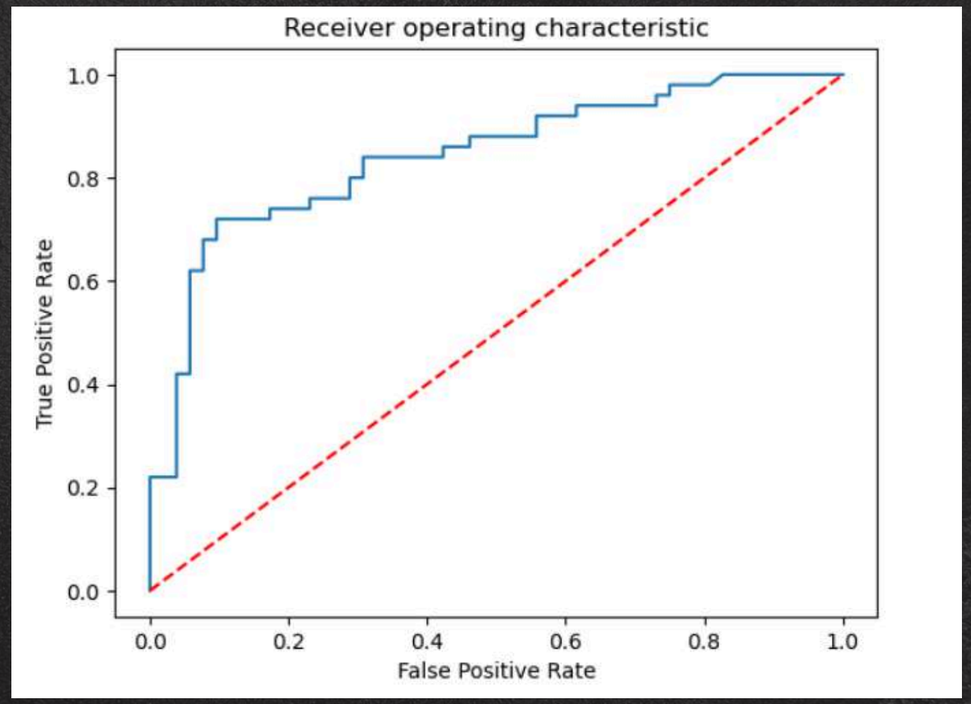
```
[[29  7]
 [ 2  7]]
```

```
Accuracy_test = ((29+7)/(45)*100)
Accuracy_test
```

```
80.0
```

Model Training Using Logistic Regression

Step 9 : ROC Curve



- The model performs significantly better than random guessing (it is well above the random classifier line).
- It can correctly identify a high proportion of true positive cases while keeping false alarms relatively low.
- The curve's shape suggests strong discriminatory power, likely in the "good" performance range (0.75–0.80).

Conclusion

From this analysis below are the major takeaway :

- Quick Commerce is bringing a Structural Shift in customer shopping behavior.
- This shift is mainly affected by Convenience of Home Delivery of goods as and when required , Offers & Discounts , Speed of Delivery , Easy to use interface.
- Traditional retail maintains higher perceived product quality trust for bulk shoppers although it is not enough to compete with Quick commerce.
- Consumers often exhibit higher average spending on quick commerce platforms compared to traditional retail channels.
- Retail is stronger when it comes to experience, trust, buying things immediately, and building personal relationships with customers while Quick commerce is stronger in fast delivery and convenience. Both options meet different customer needs. Consumer preference is influenced by the nature of the product, the customer’s profile, and the specific purchasing context.

Thank you!